

Chungbuk National University
College of Electronics & Information Engineering

VEHICULAR WIRELESS COMMUNICATION

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REVISION EXAM

Instructor: Asst. Prof. Shimaa Kotb Abdelnabi, Ph.D.
Electronic Engineering Department · CBNU

Lecture 1 · Vehicular Communication Basics

1. What fundamentally distinguishes a VANET from a general MANET?
 - A) VANETs require internet access at all times
 - B) VANETs operate in vehicular environments with very high mobility, frequent topology changes, and safety-critical constraints**
 - C) VANETs use only infrastructure-based communication
 - D) VANETs are static networks with fixed routing tables

✓ Correct Answer: B
2. In a VANET, if RSUs are removed entirely, which statement best describes the remaining network?
 - A) The network ceases to function
 - B) Vehicles can still communicate in pure V2V ad hoc mode using OBUs**
 - C) Cellular coverage automatically replaces RSU functionality
 - D) Vehicles must wait for RSUs to be restored before communicating

✓ Correct Answer: B
3. Which statement correctly describes CSMA/CA compared to CSMA/CD?
 - A) CSMA/CD is used in wireless networks; CSMA/CA is used in wired networks
 - B) CSMA/CA avoids collisions before transmission using backoff; CSMA/CD detects collisions after they occur**
 - C) CSMA/CA is faster than CSMA/CD for vehicular use
 - D) CSMA/CD is preferred for DSRC because it has lower overhead

✓ Correct Answer: B
4. What is the primary reason multi-hop packet forwarding is essential in VANETs?
 - A) Single-hop transmission always reaches all vehicles
 - B) It extends communication range beyond a single node's direct radio reach without requiring infrastructure**
 - C) It reduces the need for GNSS modules
 - D) It simplifies routing protocols by eliminating the need for certificates

✓ Correct Answer: B
5. Which statement about OBU components is CORRECT?
 - A) The GNSS module controls braking
 - B) The security module is optional in modern OBUs
 - C) The V2X communication module handles both DSRC and C-V2X protocols**
 - D) The antenna system is only used for cellular communication

✓ Correct Answer: C
6. A connected vehicle provides '360-degree situational awareness' — what technology primarily enables this compared to automated vehicles?
 - A) Higher resolution cameras
 - B) V2X communication systems that share data from vehicles and infrastructure beyond sensor line-of-sight**
 - C) Better LIDAR resolution
 - D) Faster in-vehicle processors

✓ Correct Answer: B

7. Why does V2V communication detect driver intent (e.g., braking) more reliably than camera-based systems?

A) Cameras cannot detect color

B) BSMs transmit brake status, speed, and trajectory data directly — information cameras can only infer indirectly

C) V2V has higher resolution than cameras

D) V2V operates at night while cameras do not

✓ Correct Answer: B

8. Which property makes ad hoc networks 'self-configuring'?

A) They rely on a central server to assign IP addresses

B) Nodes discover each other and establish routing paths dynamically without pre-configuration

C) Network administrators configure them before deployment

D) They use static IP addresses assigned at manufacturing

✓ Correct Answer: B

9. In the V2X taxonomy, V2P communication is most critical for:

A) Reducing highway congestion

B) Warning pedestrians and cyclists of approaching vehicles at intersections or crossings

C) Providing cloud-based navigation updates

D) Enabling vehicle platooning on motorways

✓ Correct Answer: B

10. An RSU does NOT enforce a strict hierarchical structure. What does this mean for VANET operations?

A) RSUs are the primary control nodes of all V2X traffic

B) Vehicles remain autonomous and can communicate directly without RSU involvement, preserving the ad hoc nature

C) RSUs must authenticate every V2V message

D) VANETs cannot function without at least one RSU per kilometer

✓ Correct Answer: B

11. What is the main technical reason why V2V is categorized as 'platooning' (V2V use case)?

A) Platooning requires cellular connectivity through the Uu interface

B) Platooning requires direct real-time vehicle-to-vehicle exchange of speed, position, and braking data to maintain tight formation

C) Platooning works best with RSU coordination only

D) Platooning is a V2N application because it requires cloud coordination

✓ Correct Answer: B

12. What does 'peer-to-peer' communication in MANETs mean for routing responsibilities?

A) One designated master node handles all routing

B) Every node acts as both transmitter and router, forwarding packets on behalf of others

C) Only RSUs forward packets between vehicles

D) Peer-to-peer means direct one-hop communication only — no forwarding

✓ Correct Answer: B

13. Which of these is the most technically accurate reason RSUs are mounted with omnidirectional antennas?

- A) To reduce interference between adjacent RSUs
- B) To broadcast coverage equally in all 360° directions so vehicles approaching from any direction receive messages**
- C) To increase range beyond 5 km
- D) Omnidirectional antennas are cheaper than directional ones

✓ Correct Answer: B

14. A mesh network node fails. Which behavior characterizes a self-healing mesh network?

- A) The network stops working until the node is replaced
- B) Traffic is automatically rerouted through alternative paths without human intervention**
- C) A central server assigns new paths to all nodes
- D) Adjacent nodes must be manually reconfigured

✓ Correct Answer: B

15. Which statement about the OBU GNSS module is most technically precise?

- A) GNSS provides connectivity to cellular networks
- B) GNSS provides precise vehicle position data included in BSMs for collision detection algorithms**
- C) GNSS manages V2X security certificates
- D) GNSS communicates directly with RSUs to relay position data

✓ Correct Answer: B

16. Why are BSMs broadcast 10 times per second specifically?

- A) 10 Hz is the maximum rate supported by IEEE 802.11p
- B) At typical highway speeds, 100ms intervals provide sufficient position update frequency for collision avoidance algorithms**
- C) 10 Hz matches the refresh rate of vehicle dashboards
- D) The 5.9 GHz spectrum is partitioned into exactly 10 slots per second

✓ Correct Answer: B

17. What does 'dynamic topology' in MANETs/VANETs mean for routing protocols?

- A) Routes are pre-computed and stored in each vehicle
- B) Routes change frequently as vehicles move, requiring protocols that can discover and update paths on-the-fly**
- C) Topology is managed centrally by the TMC
- D) Dynamic topology only applies to infrastructure-based networks

✓ Correct Answer: B

18. Which factor makes VANETs more challenging than general MANETs from a network design perspective?

- A) VANETs use more expensive hardware
- B) Vehicles move much faster and more predictably (road-constrained) but with very short contact durations between nodes**
- C) VANETs require more frequency bands than MANETs
- D) VANETs have smaller network sizes than MANETs

✓ Correct Answer: B

19. The 'no central infrastructure' property of ad hoc networks creates which specific challenge for VANETs?

A) Lack of GPS support

B) No centralized trust anchor for authentication — requiring distributed security mechanisms like SCMS

C) Vehicles cannot use OFDM modulation

D) Channel access cannot be coordinated at all

✓ Correct Answer: B

20. If a vehicle with only an OBU (no RSU nearby) needs to warn a vehicle 800m ahead about a hazard, what must occur?

A) The warning cannot be sent as DSRC range is 300m

B) Multi-hop relaying — intermediate vehicles forward the BSM extending the effective range

C) The vehicle must use cellular V2N to send the alert

D) No mechanism exists for this scenario in VANET design

✓ Correct Answer: B

Lecture 2 · Connected Vehicles and V2I Communication

21. Why is V2I communication at tunnels particularly valuable compared to open roads?
- A) Tunnels have better cellular coverage
 - B) Tunnels block line-of-sight for sensors and GPS signals, making infrastructure-provided ahead-of-curve warnings critical**
 - C) RSUs in tunnels use higher power
 - D) V2V works better inside tunnels due to signal reflection
- ✓ Correct Answer: B
22. A connected vehicle receives a SPaT message. What does the vehicle do with this data?
- A) It updates its infotainment map display
 - B) It uses signal phase and timing to optimize speed for green light passage or prepare to stop**
 - C) It reports the data to the TMC for logging
 - D) It re-broadcasts the SPaT to nearby vehicles via V2V
- ✓ Correct Answer: B
23. What distinguishes a 'Connected Automated Vehicle' (CAV) from a standard 'Automated Vehicle' (AV)?
- A) CAVs have more sensors than AVs
 - B) CAVs combine internal sensing (automation) with external V2X communication for information beyond sensor range**
 - C) AVs use cellular networks while CAVs use DSRC
 - D) CAVs do not require any internal sensors
- ✓ Correct Answer: B
24. BSMs are broadcast without acknowledgment. What does this 'broadcast-only' nature imply for system design?
- A) Every vehicle must confirm receipt of each BSM
 - B) Safety applications must tolerate occasional packet loss and use redundant periodic transmission to compensate**
 - C) BSMs are encrypted point-to-point to the nearest RSU
 - D) TCP is used to guarantee delivery of each BSM
- ✓ Correct Answer: B
25. Why is cooperative mapping preferable to individual vehicle mapping for localization accuracy?
- A) Individual vehicles have more processing power
 - B) Multiple vehicles contribute overlapping sensor data from different angles, reducing cumulative localization error**
 - C) Cooperative maps use satellite data only
 - D) Individual mapping is more accurate because it avoids data conflicts
- ✓ Correct Answer: B
26. In a V2I system, what is the functional difference between a TMC and an RSU?
- A) TMC is a physical roadside device; RSU is a remote data center
 - B) RSU handles local wireless V2X communication; TMC is a centralized system coordinating traffic management using remote communications**
 - C) TMC broadcasts safety messages; RSU manages cellular handover

D) Both perform identical functions at different geographic scales

✓ **Correct Answer: B**

27. Which scenario demonstrates the limitations of radar compared to V2V communication?

A) Detecting a vehicle directly in front at 10m

B) Detecting brake lights of a vehicle two cars ahead blocked by a truck (non-line-of-sight)

C) Measuring precise vehicle speed

D) Detecting lane markings on wet roads

✓ **Correct Answer: B**

28. A V2I system uses PSID (Provider Service Identifier). What is the technical purpose of PSID?

A) It identifies the vehicle's license plate over V2X

B) It identifies which application a V2X message belongs to, allowing devices to filter relevant messages

C) It encrypts the V2I message payload

D) It assigns a radio channel for the V2I communication

✓ **Correct Answer: B**

29. Connectivity extends the sensing range of vehicles. What specific technical mechanism enables this?

A) Better antenna gain

B) Vehicles share their sensor perceptions (position, speed, hazard data) via V2X, creating a collective sensing range larger than any individual sensor

C) RSUs amplify radar signals from vehicles

D) Cellular networks relay sensor data between vehicles

✓ **Correct Answer: B**

30. In rural overtaking scenarios (passing on rural roads), why is V2V specifically more valuable than radar?

A) Radar is illegal on rural roads

B) Oncoming vehicles beyond the visual horizon can proactively share position/speed via BSM, while radar requires line-of-sight

C) V2V operates at longer range frequencies on rural roads

D) Rural roads have more RSUs than urban roads

✓ **Correct Answer: B**

31. What is the technical significance of OBU processing BSMs '10 times per second' for safety applications?

A) It matches the GPS update rate of 10 Hz

B) At 100km/h, a vehicle travels 2.8m per 100ms — frequent updates are needed to maintain accurate neighboring vehicle models for collision assessment

C) 10 Hz matches the human reaction time threshold

D) IEEE 1609.3 mandates exactly 10 Hz for all V2X messages

✓ **Correct Answer: B**

32. A connected vehicle loses RSU connectivity temporarily. Which statement is MOST accurate?

A) All safety functions cease immediately

B) V2V communication continues unaffected — V2I loss degrades infrastructure-based services but not peer-to-peer safety messaging

C) The vehicle automatically switches to satellite

D) BSM transmission automatically stops without RSU coverage

✓ **Correct Answer: B**

33. What does 'infrastructure sensing' provide that individual vehicle sensing cannot?

A) Better camera resolution

B) A fixed elevated vantage point that can monitor traffic without being blocked by other vehicles, providing persistent coverage

C) Lower latency than in-vehicle sensors

D) Longer range radar than OBU-mounted systems

✓ **Correct Answer: B**

34. The display panel in a connected vehicle receives an alert. What determines whether the alert is shown or acted upon automatically?

A) The RSU decides which alerts to show

B) The OBU application layer evaluates threat severity and context to determine whether to warn the driver or trigger an automated response

C) The TMC sends display instructions to each vehicle individually

D) All received BSMs trigger dashboard alerts automatically

✓ **Correct Answer: B**

35. Why does removing traffic lights at intersections become feasible with V2I connected infrastructure?

A) Cameras on RSUs replace signal timing

B) Vehicles can communicate position and intent via V2X, allowing real-time cooperative negotiation of intersection passage without fixed signal phases

C) RSUs replace physical traffic lights with audio signals

D) Connected vehicles simply ignore traffic lights

✓ **Correct Answer: B**

36. What does 'path history' in a BSM enable receiving vehicles to calculate?

A) The sender vehicle's fuel consumption

B) A trajectory prediction of the sending vehicle's future path for proactive collision threat assessment

C) The insurance status of the sending vehicle

D) The sending vehicle's destination for routing purposes

✓ **Correct Answer: B**

37. Why is V2I communication characterized as 'generally broadcast with some two-way transactions'?

A) RSUs cannot receive V2V messages

B) Most V2I is one-way broadcast (RSU → vehicles) but some services like tolling require bidirectional data exchange

C) Two-way transactions require cellular networks, not DSRC

D) Broadcast-only V2I is mandated by IEEE 1609.4

✓ **Correct Answer: B**

38. How does V2X connectivity help vehicles 'with different automation levels' interact?

A) Highly automated vehicles override lower-automation ones

B) V2X provides a common communication layer — a fully automated vehicle can receive intent/position from a manually driven vehicle that has an OBU

- C) Automation level is automatically detected by RSUs
- D) Vehicles with different automation levels cannot share V2X messages

✓ **Correct Answer: B**

39. In V2I, what is the purpose of 'Vehicle Data' (latitude, longitude, speed, brake status) sent TO infrastructure?

- A) The TMC uses it to control the vehicle remotely
- B) Infrastructure aggregates vehicle probe data to monitor traffic conditions and improve dynamic signal timing and hazard detection**
- C) The RSU retransmits vehicle data back to the same vehicle
- D) Vehicle data is stored in OBU memory only and never transmitted

✓ **Correct Answer: B**

40. Which metric is MOST directly linked to V2I system effectiveness at preventing intersection crashes?

- A) Total number of RSUs deployed
- B) End-to-end latency of SPaT and MAP messages — lower latency means vehicles receive signal timing in time to adjust speed appropriately**
- C) Channel bandwidth available to the RSU
- D) Number of vehicles with infotainment displays

✓ **Correct Answer: B**

Lecture 3 · DSRC Technology and WAVE Architecture

41. DSRC uses IEEE 802.11p (not standard Wi-Fi). What specific modification makes 802.11p suitable for vehicular use?

- A) It uses a different frequency band
- B) It eliminates the association/handshake process enabling immediate one-way broadcast — critical for vehicles passing at high speed**
- C) It increases channel bandwidth to 40 MHz
- D) It adds TCP reliability to DSRC transmissions

✓ Correct Answer: B

42. The DSRC protocol stack places WSMP alongside IP/UDP. What does this architectural choice enable?

- A) WSMP provides a backup path if IP fails
- B) Safety-critical BSMs bypass the heavier IP stack via WSMP for lower latency, while non-safety apps use IP/UDP**
- C) WSMP encrypts IP packets before transmission
- D) WSMP is only used for RSU-to-TMC communication

✓ Correct Answer: B

43. IEEE 1609.4 defines multi-channel operation. What specific problem does this solve in DSRC?

- A) It eliminates the need for a control channel
- B) It allows devices to alternate between the control channel (safety) and service channels (non-safety), preventing safety messages from being displaced by non-safety traffic**
- C) It allows all 7 channels to be used simultaneously
- D) It solves the hidden node problem in dense deployments

✓ Correct Answer: B

44. DSRC uses OFDM modulation. Why is OFDM specifically chosen over simpler modulations for vehicular channels?

- A) OFDM requires less processing power
- B) OFDM is robust against multipath fading and Doppler spread caused by fast-moving vehicles — sub-carriers spread the signal across frequency**
- C) OFDM provides higher power output
- D) OFDM eliminates the need for error correction codes

✓ Correct Answer: B

45. The DSRC control channel (CH 178) is separated from service channels. Why must safety messages ONLY use the control channel?

- A) Service channels do not support BSM format
- B) Guaranteed airtime on the control channel ensures safety messages are not delayed by competing non-safety service channel traffic**
- C) The control channel has higher transmit power
- D) Service channels are encrypted and cannot carry unencrypted BSMs

✓ Correct Answer: B

46. DSRC has a 10 MHz channel bandwidth — half of standard Wi-Fi (20 MHz). What is the technical consequence?

A) DSRC has twice the data rate of Wi-Fi

B) Lower bandwidth means lower data rate but better coverage and robustness in high-mobility, high-Doppler vehicular environments

C) DSRC cannot carry safety messages due to low bandwidth

D) 10 MHz allows DSRC to operate at longer range than Wi-Fi

✓ **Correct Answer: B**

47. IEEE 1609.2 handles security in DSRC. What specific security service does it provide for BSMs?

A) End-to-end encryption of BSM payload

B) Digital signatures using pseudonym certificates to authenticate the message sender without revealing permanent identity

C) Firewall functions for RSU-to-vehicle communication

D) Symmetric key encryption for all V2V messages

✓ **Correct Answer: B**

48. DSRC operates in the 5.9 GHz band. What regulatory reason specifically reserves this spectrum for V2X?

A) 5.9 GHz provides the best range for vehicular communication

B) The 75 MHz allocation (5.850–5.925 GHz) was specifically licensed for safety-critical ITS use, isolating it from general consumer Wi-Fi interference

C) 5.9 GHz is cheaper to license than other bands

D) 5.9 GHz was chosen because it has no existing users

✓ **Correct Answer: B**

49. WSMP (WAVE Short Message Protocol) is used instead of TCP for safety messages. Why is TCP unsuitable for V2V safety messaging?

A) TCP requires more bandwidth than DSRC can provide

B) TCP's handshake, acknowledgment, and retransmission overhead add latency incompatible with sub-50ms safety alert requirements

C) TCP cannot operate in the 5.9 GHz band

D) TCP is not supported in IEEE 802.11p at all

✓ **Correct Answer: B**

50. CSMA/CA is the MAC layer for DSRC. What specific mechanism does CSMA/CA use to reduce collision probability?

A) It detects collisions after they occur and retransmits immediately

B) Each node listens for channel activity and, if busy, waits a random backoff period before attempting transmission

C) CSMA/CA uses time slots assigned by an RSU

D) CSMA/CA prioritizes messages by vehicle speed

✓ **Correct Answer: B**

51. WAVE Service Advertisement (WSA) is defined in IEEE 1609.3. What is its function?

A) It advertises available road services to drivers via dashboard display

B) It allows service providers to announce available services and parameters, enabling devices to discover and join service channels

C) It is a security advertisement broadcast by SCMS

D) WSA is used only for toll payment services

✓ **Correct Answer: B**

52. The DSRC spectrum allocates CH 172 and CH 184 as 'Service channels (safety only)'. What distinguishes these from regular service channels?

A) They operate at higher power

B) They are reserved for safety-only service communications, not general non-safety applications, maintaining safety message prioritization

C) They use a different modulation scheme

D) They are shared with DSRC's control channel

✓ **Correct Answer: B**

53. Why does DSRC specify a maximum of 10 MHz per channel rather than using wider channels?

A) Wider channels are not supported by IEEE 802.11

B) Narrower channels improve robustness under Doppler shift at high vehicle speeds and allow more channels within the 75 MHz allocation

C) 10 MHz provides the highest data rate for V2X

D) Regulatory bodies mandated 10 MHz before the standard was written

✓ **Correct Answer: B**

54. What is the purpose of the 'Reserved 5 MHz' at the bottom of the DSRC band?

A) It is used for RSU-to-TMC backhaul

B) It acts as a guard band to prevent interference with adjacent spectrum allocations below 5.850 GHz

C) It is reserved for future 5G deployment

D) Emergency vehicles use this 5 MHz exclusively

✓ **Correct Answer: B**

55. DSRC authentication uses pseudonym certificates that change frequently. Why would a vehicle using the SAME certificate for months create a problem?

A) Certificates expire after 30 days regardless

B) A stable long-term identifier allows observers to link all messages to one vehicle, enabling tracking and privacy violation

C) Long-lived certificates are computationally expensive to verify

D) The SCMS mandates certificate changes for billing purposes

✓ **Correct Answer: B**

56. In DSRC, what happens when a vehicle moves out of RSU coverage but continues driving?

A) All V2X communication stops

B) The vehicle continues V2V communication via PC5 — RSU is only needed for V2I services, not V2V safety messages

C) The vehicle automatically switches to 4G LTE for V2V

D) The OBU enters sleep mode to conserve power

✓ **Correct Answer: B**

57. IEEE 1609.4 requires devices to switch between CCH and SCH in interleaved intervals. What latency concern does this create?

A) It eliminates all latency for safety messages

B) If a BSM arrives during a SCH interval, it must wait until the next CCH interval — potentially adding 50ms+ delay

C) Channel switching takes 0ms in 802.11p hardware

D) Safety messages always interrupt SCH to avoid delay

✓ **Correct Answer: B**

58. DSRC's BSM is broadcast without acknowledgment. Which design principle justifies this decision?

A) Acknowledgment reduces latency

B) In V2V safety messaging, the overhead of unicast acknowledgment from hundreds of neighbors would overwhelm the channel — reliable delivery relies on high-frequency repetition instead

C) IEEE 802.11p does not support acknowledgment frames

D) Acknowledgment is handled by the application layer in DSRC

✓ **Correct Answer: B**

59. The DSRC standard separates IEEE 802.11p (PHY/MAC) from IEEE 1609.x (upper layers). What architectural benefit does this separation provide?

A) It allows different modulation schemes per 1609.x standard

B) It modularizes the stack — upper-layer protocols (security, networking, applications) can evolve independently without changing the radio hardware

C) It reduces the cost of OBU manufacturing

D) Separation forces all V2X to use the same frequency

✓ **Correct Answer: B**

60. Why is DSRC described as having 'limited scalability in dense environments' despite using CSMA/CA?

A) Dense environments have too many RSUs

B) As vehicle density increases, more nodes compete for the shared channel simultaneously, increasing collision probability and channel load despite CSMA/CA's backoff mechanism

C) DSRC uses a fixed number of time slots that cannot scale

D) Dense environments cause GNSS interference with DSRC

✓ **Correct Answer: B**

Lecture 4 · DSRC Requirements, Metrics, and Challenges

61. Why is 'Packet Reception Ratio' (PRR) a more meaningful metric than throughput for evaluating DSRC safety systems?

A) PRR is easier to measure than throughput

B) Safety depends on what fraction of transmitted safety messages are successfully received — a high throughput system with low PRR may still fail to prevent collisions

C) Throughput cannot be measured in broadcast systems

D) PRR is mandated by IEEE standards while throughput is not

✓ **Correct Answer: B**

62. At 250 km/h, two vehicles approach each other head-on. What specific DSRC design challenge does this extreme speed create?

A) Channel bandwidth is insufficient

B) Doppler spread is very high, degrading OFDM subcarrier orthogonality — requiring specific Doppler compensation in the PHY layer

C) BSMs cannot be transmitted at vehicle speeds above 200 km/h

D) GPS accuracy degrades at speeds above 200 km/h

✓ **Correct Answer: B**

63. 'Hidden node' scenarios reduce DSRC reliability. What is the specific mechanism of failure?

A) RSUs block line-of-sight between vehicles

B) Vehicle A and Vehicle C cannot detect each other's transmissions but both transmit to Vehicle B simultaneously, causing a collision at B that neither A nor C detects

C) GNSS errors cause incorrect position reporting

D) Encryption overhead delays transmission enough to cause timing collisions

✓ **Correct Answer: B**

64. Adaptive message rate control (like LIMERIC) reduces channel load. What unintended consequence must system designers consider?

A) Lower message rate increases encryption overhead

B) Reducing beacon rate in high-density scenarios means position updates are less frequent, potentially degrading tracking accuracy for fast-moving vehicles

C) Adaptive rate control increases RSU processing load

D) Lower message rates reduce GPS power consumption

✓ **Correct Answer: B**

65. DSRC channel congestion is specifically problematic because:

A) Congestion causes RSUs to shut down

B) High channel load degrades exactly the safety messages that high-traffic (high-density) scenarios most urgently require — congestion peaks when safety is most critical

C) Congestion only affects non-safety service channels

D) Congestion increases latency for cellular V2N services

✓ **Correct Answer: B**

66. Why is the 5.9 GHz sharing with Wi-Fi a technically complex problem beyond simple interference?

A) Wi-Fi uses more power than DSRC

B) Wi-Fi devices use aggressive CSMA/CA without DSRC-specific prioritization, potentially disrupting safety messages and fragmenting the band if DSRC and Wi-Fi must coexist

C) 5.9 GHz Wi-Fi is illegal in all V2X deployment regions

D) Wi-Fi and DSRC use different modulation schemes incompatible at the physical layer

✓ **Correct Answer: B**

67. What is the technical root cause of the 'service channel switching delay' problem in DSRC?

A) RSUs do not synchronize their channel switching

B) IEEE 1609.4 mandates alternating CCH and SCH intervals — if a safety event occurs during an SCH interval, the BSM must wait until the next CCH interval to be transmitted

C) The OBU hardware requires 10ms to physically change frequency

D) Service channel switching only affects non-safety applications

✓ **Correct Answer: B**

68. In DSRC, why is 'adaptive transmit power control' an alternative (or complement) to message rate control for congestion management?

A) Higher power increases range and helps dense networks

B) Reducing transmit power limits the radio range — fewer vehicles 'hear' each transmission, reducing contention on the shared channel

C) Adaptive power control is mandated by IEEE 1609.2

D) Power control increases PRR by boosting signal strength

✓ **Correct Answer: B**

69. DSRC requires low-latency communication. Which latency budget breakdown is most technically accurate for a V2V safety alert?

A) All latency is from propagation delay

B) Total latency includes: channel access wait time (CSMA/CA backoff) + transmission time + processing time — backoff can dominate under congestion

C) Latency is fixed at exactly 50ms by IEEE 802.11p

D) Latency only matters for V2I applications, not V2V

✓ **Correct Answer: B**

70. Why is 'line-of-sight (LOS)' degradation in urban environments a fundamental DSRC challenge rather than a simple engineering problem?

A) Urban buildings reflect signals, improving range

B) Buildings, trucks, and obstacles create non-line-of-sight (NLOS) conditions that introduce multipath fading and can block safety-critical messages at the exact locations (intersections) where they are most needed

C) Urban environments have better RSU coverage to compensate

D) DSRC antennas in urban environments can be raised to overcome building blockage

✓ **Correct Answer: B**

71. What specific technical problem does 'message filtering and prioritization' solve in dense DSRC networks?

A) It reduces the number of RSUs needed

B) It selectively forwards only the most relevant messages (e.g., nearby vehicles) and assigns higher priority to safety vs non-safety, reducing effective channel load without reducing safety message delivery

- C) Message filtering is applied at the RSU level only
- D) Prioritization ensures all 7 DSRC channels are used equally

✓ **Correct Answer: B**

72. PRR decreases with increasing vehicle density even with CSMA/CA. Why does CSMA/CA fail to fully protect PRR?

- A) CSMA/CA cannot detect transmissions from more than 5 neighbors
- B) As density grows, backoff periods become insufficient — multiple nodes simultaneously exit backoff and transmit, causing collisions that CSMA/CA cannot prevent (only reduce)**
- C) CSMA/CA was not designed for vehicular environments
- D) High density causes GNSS errors that corrupt BSMs

✓ **Correct Answer: B**

73. 'Timing synchronization' is important in DSRC. What specific failure occurs without precise time synchronization?

- A) GPS position data becomes inaccurate
- B) CCH/SCH channel switching in IEEE 1609.4 requires synchronized timeslots across all devices — without sync, devices switch channels at different times and miss messages from neighbors**
- C) BSM transmission rate drops below 10 Hz
- D) Security certificates cannot be validated without synchronized clocks

✓ **Correct Answer: B**

74. DSRC prioritizes safety messaging over high bandwidth. What specifically is sacrificed to achieve low latency?

- A) Message range is reduced
- B) Data rate is limited (3–27 Mbps) and channel bandwidth is narrow (10 MHz) to optimize for robustness and low latency at the cost of throughput**
- C) Security services are disabled
- D) V2I communication is disabled in safety priority mode

✓ **Correct Answer: B**

75. A DSRC system with high PRR but high latency would still fail safety requirements. Why?

- A) High latency means RSUs reject the message
- B) A collision warning received 300ms after the collision trigger point provides insufficient time for driver response or automated braking — timing and reliability must both be met**
- C) High latency automatically reduces PRR
- D) DSRC hardware cannot operate with latency above 100ms

✓ **Correct Answer: B**

76. Why is 'channel load' the central design variable in DSRC congestion control rather than simply limiting vehicle count?

- A) Vehicle count cannot be controlled by communication systems
- B) Channel load directly determines PRR and latency — it can be controlled by adjusting beacon rate and power per vehicle, regardless of total vehicle count**
- C) Channel load and vehicle count are always equal
- D) Limiting vehicle count is the primary solution in current standards

✓ **Correct Answer: B**

77. What specific antenna-related challenge affects DSRC communication in truck platooning scenarios?

A) Trucks use different frequency bands

B) Trucks obstruct radio signals between vehicles in different lanes or behind them (NLOS blockage), requiring careful antenna placement on the vehicle to maintain LOS to nearby OBUs

C) Platooning requires directional antennas not supported by DSRC

D) Truck-mounted OBUs use higher power automatically

✓ **Correct Answer: B**

78. Why are event-driven BSMs (e.g., emergency brake) treated differently from periodic BSMs in DSRC system design?

A) Event-driven BSMs use a different certificate

B) Event-driven BSMs require immediate transmission with higher priority — they cannot wait for the next scheduled periodic slot as the safety event may resolve in milliseconds

C) Event-driven BSMs are transmitted on service channels

D) Periodic and event-driven BSMs are identical in DSRC — no distinction is made

✓ **Correct Answer: B**

79. A low 'Packet Reception Ratio' in DSRC is observed at an intersection. Which combination of factors most likely explains this?

A) Too few vehicles and high RSU density

B) High vehicle density creating channel congestion + building blockage creating NLOS conditions — both factors simultaneously degrade PRR at intersections

C) Insufficient transmit power and low vehicle density

D) Security certificate overhead consuming channel capacity

✓ **Correct Answer: B**

80. DSRC's reliability is evaluated 'together with latency rather than throughput.' What does this reveal about DSRC's fundamental design philosophy?

A) DSRC is optimized for maximum data throughput

B) DSRC is designed for time-critical safety message delivery where both the probability of reception AND the timeliness of reception jointly determine whether a safety warning is effective

C) Throughput is irrelevant in all V2X systems

D) Latency in DSRC is always zero due to direct radio transmission

✓ **Correct Answer: B**

Lecture 5 · Introduction to Cellular V2X (C-V2X)

81. Why does C-V2X PC5 Mode 4 use 'sensing-based' resource selection rather than random selection?
- A) Random selection is prohibited by 3GPP Release 14
 - B) Sensing allows the UE to monitor occupied resources and select unoccupied ones, reducing collision probability compared to random selection**
 - C) Mode 4 sensing requires RSU coordination
 - D) Sensing-based selection reduces Doppler shift effects
- ✓ Correct Answer: B
82. LTE-V2X was introduced in 3GPP Release 14. Why was it important that it evolved from LTE rather than being built from scratch?
- A) New standards take longer to develop than LTE extensions
 - B) Leveraging LTE's existing infrastructure, chipset ecosystem, frequency bands, and operator relationships dramatically accelerated deployment and reduced cost**
 - C) LTE has better modulation than DSRC
 - D) 3GPP mandated that all V2X must be LTE-based
- ✓ Correct Answer: B
83. PC5 operates in the 5.9 GHz ITS band. What does this mean for its relationship to DSRC spectrum?
- A) PC5 and DSRC share exactly the same channels, creating guaranteed interference
 - B) PC5 uses the same 5.9 GHz ITS band allocation as DSRC — in deployments requiring coexistence, spectrum sharing between C-V2X and DSRC becomes a critical technical and regulatory challenge**
 - C) PC5 uses a different part of the 5.9 GHz band with no overlap
 - D) PC5 operates at 3.5 GHz, not 5.9 GHz
- ✓ Correct Answer: B
84. C-V2X latency targets are 'under 10ms.' Achieving this requires what specific architectural choice?
- A) Always routing messages through the eNodeB
 - B) Using PC5 direct communication — routing through eNodeB (Uu) adds network delays that exceed the 10ms target for safety-critical applications**
 - C) Using TCP for reliable delivery
 - D) Increasing transmission power to reduce propagation delay
- ✓ Correct Answer: B
85. Mode 3 vs Mode 4 in LTE-V2X PC5 — which mode is MOST appropriate for a vehicle in a remote rural area with no cellular coverage?
- A) Mode 3, because the eNodeB coordinates rural traffic
 - B) Mode 4, because it requires no network infrastructure — vehicles autonomously select resources through sensing**
 - C) Both modes work equally well without cellular coverage
 - D) Neither mode works without cellular coverage in LTE-V2X
- ✓ Correct Answer: B
86. The Uu interface provides 'cloud integration.' What specific V2X use case benefits MOST from cloud integration via Uu?
- A) Emergency braking between adjacent vehicles

B) HD map updates and wide-area traffic coordination — services requiring large data volumes or aggregating information from many vehicles across a region

C) Collision avoidance at 10m distance

D) Pedestrian alerts at crosswalks

✓ **Correct Answer: B**

87. C-V2X supports OFDM as its multiplexing scheme. What advantage does OFDM specifically provide for C-V2X in vehicular environments?

A) OFDM simplifies security certificate management

B) OFDM's multi-subcarrier structure provides robustness against frequency-selective fading and Doppler spread caused by vehicle mobility

C) OFDM allows C-V2X to operate without a GNSS module

D) OFDM is required because the 5.9 GHz band mandates it

✓ **Correct Answer: B**

88. What does 'Semi-Persistent Scheduling' (SPS) in Mode 4 specifically mean?

A) Resources change randomly every BSM transmission

B) The UE semi-persistently occupies selected resources for multiple consecutive BSM transmissions, with periodic re-evaluation — reducing overhead compared to fully dynamic selection

C) The eNodeB pre-assigns resources that never change

D) SPS only applies to V2N communication via the Uu interface

✓ **Correct Answer: B**

89. C-V2X improved upon DSRC in 'range, reliability, and congestion handling.' What specific technical mechanism gives C-V2X better congestion handling?

A) C-V2X uses more channels than DSRC

B) C-V2X's sensing-based resource allocation (Mode 4) and network-coordinated scheduling (Mode 3) are more sophisticated than DSRC's CSMA/CA, reducing resource collision probability

C) C-V2X has higher transmit power limits

D) C-V2X uses TCP to guarantee delivery

✓ **Correct Answer: B**

90. In C-V2X, the evolution from LTE-V2X (Release 14) toward 5G NR-V2X was driven by which fundamental technical gap?

A) LTE-V2X could not support BSM transmission

B) LTE-V2X latency (20ms) and reliability (>90%) were insufficient for cooperative autonomous driving which requires <1ms latency and 99.999% reliability

C) LTE-V2X could not operate in the 5.9 GHz band

D) LTE-V2X did not support the PC5 interface

✓ **Correct Answer: B**

91. Why can PC5 direct communication operate 'out-of-coverage' while Uu cannot?

A) PC5 uses satellite as fallback when cellular is unavailable

B) PC5 is a device-to-device direct link — it never passes through the cellular base station infrastructure, so cellular coverage is irrelevant for PC5 operation

C) Out-of-coverage PC5 uses higher power automatically

D) PC5 switches to Wi-Fi Direct when cellular coverage is lost

✓ Correct Answer: B

92. A vehicle using Mode 3 (eNodeB-scheduled) moves into a cellular coverage gap. What happens to its PC5 communication?

- A) PC5 communication stops entirely
- B) The vehicle must switch to Mode 4 (autonomous sensing-based resource selection) to continue direct V2X communication without eNodeB coordination**
- C) The vehicle automatically uses Uu to continue V2V communication
- D) Mode 3 creates a resource reservation that persists after coverage loss

✓ Correct Answer: B

93. C-V2X's 3GPP standardization specifically enables an 'evolution path to 5G.' What technically makes this evolution possible?

- A) LTE-V2X and NR-V2X share identical hardware
- B) LTE-V2X's PC5 and Uu interface concepts, resource management approaches, and protocol structures were designed to be upgradeable to 5G NR — Release 16 extends rather than replaces them**
- C) 5G uses the same modulation as LTE-V2X
- D) Evolution is possible because LTE-V2X and 5G use the same frequencies

✓ Correct Answer: B

94. Both DSRC and C-V2X PC5 operate in the 5.9 GHz band. Why is this co-deployment technically controversial?

- A) Both use identical MAC protocols that guarantee coexistence
- B) DSRC uses CSMA/CA while C-V2X PC5 does not — they have incompatible channel access mechanisms, so co-deployment in the same channels causes asymmetric interference**
- C) Regulatory bodies prohibit both in the same country
- D) DSRC uses 75 MHz while C-V2X uses 20 MHz of a different band

✓ Correct Answer: B

95. What distinguishes 'cooperative collision avoidance' (C-V2X) from traditional radar-based collision avoidance?

- A) C-V2X uses cameras while radar uses radio waves
- B) C-V2X enables proactive sharing of intent, trajectory, and braking data between cooperating vehicles — radar is reactive (measures distance to detected objects without their cooperation)**
- C) Radar operates at longer range than C-V2X
- D) C-V2X collision avoidance only works at intersections

✓ Correct Answer: B

96. The Uu interface adds latency compared to PC5. For V2N applications, why is this higher latency acceptable?

- A) V2N applications do not need low latency
- B) V2N use cases (traffic updates, HD maps, cloud analytics) are latency-tolerant — their utility does not degrade significantly with 50-200ms delays, unlike safety-critical V2V applications**
- C) Uu latency is identical to PC5 latency
- D) V2N applications use edge computing to eliminate Uu latency

✓ Correct Answer: B

97. In C-V2X, 'reuse of DSRC/C-ITS higher layers' means what specifically?

A) C-V2X uses the same PHY/MAC as DSRC

B) Application-layer message sets (like SAE J2735 BSMs) and higher-layer protocols developed for DSRC can run over C-V2X without modification — only the lower radio layers change

C) C-V2X hardware is backward compatible with DSRC certificates

D) DSRC and C-V2X share the same security credential management system

✓ **Correct Answer: B**

98. Why is 'high density support' specifically listed as an LTE-V2X advantage over DSRC?

A) LTE-V2X uses more channels than DSRC

B) C-V2X's resource allocation (sensing/scheduling) is more collision-resistant than DSMA/CA under high vehicle densities, maintaining acceptable PRR where CSMA/CA degrades

C) LTE-V2X uses higher power in dense environments

D) LTE-V2X automatically limits the number of connected vehicles

✓ **Correct Answer: B**

99. A C-V2X system using Mode 4 on PC5 — which statement about security is most accurate?

A) Mode 4 operation eliminates the need for security certificates

B) Mode 4 still requires vehicles to use pseudonym certificates from the SCMS to sign BSMs — the resource allocation mode is independent of the security layer

C) Mode 4 uses symmetric encryption instead of certificates

D) Out-of-coverage Mode 4 operation uses pre-shared keys not from SCMS

✓ **Correct Answer: B**

100. Why does C-V2X specifically support both V2V (PC5) AND V2N (Uu) simultaneously?

A) Regulatory requirements mandate both interfaces

B) Safety-critical local interactions (V2V via PC5) and cloud-based services (V2N via Uu) have fundamentally different latency/range requirements that a single interface cannot satisfy — dual interfaces address both

C) PC5 is for downlink and Uu is for uplink

D) Simultaneous operation is required to maintain backward compatibility with DSRC

✓ **Correct Answer: B**

Lecture 6 · C-V2X Applications, Architecture, and Advantages

101. In C-V2X architecture, what specifically differentiates the 'V2X Control Function' from the 'V2X Application Server'?

- A) The Control Function is installed in vehicles; the Application Server is in RSUs
- B) The V2X Control Function performs service authorization and parameter provisioning; the V2X Application Server hosts the actual application logic — they are separate core network entities with distinct roles**
- C) Both are RSU-level components that differ only in geographic coverage
- D) The Application Server handles security; the Control Function handles routing

✓ **Correct Answer: B**

102. Network slicing in 5G-V2X creates virtual network partitions. What specific V2X problem does this solve that a single shared network cannot?

- A) It reduces the need for RSUs
- B) It allows safety-critical V2X traffic (requiring <10ms, 99.999% reliability) and infotainment traffic (tolerant of 200ms, 90% reliability) to coexist without the former being degraded by the latter**
- C) Network slicing eliminates the need for QoS classification
- D) It allows different vehicles to use different frequency bands

✓ **Correct Answer: B**

103. Why is the MAC layer specifically responsible for 'packet prioritization' in C-V2X rather than the application or IP layer?

- A) The application layer does not know message content
- B) MAC-layer prioritization acts closest to the radio channel — it directly controls which messages access the channel first, which is more effective than higher-layer queuing for real-time safety message delivery**
- C) IP-layer prioritization is more effective for V2X
- D) The security module overrides MAC-layer decisions

✓ **Correct Answer: B**

104. PC5 can operate 'without base station' — what does this specifically enable for autonomous driving in tunnels?

- A) Tunnel MEC servers replace the base station function
- B) Vehicles can continue exchanging safety messages with each other directly through the tunnel without needing cellular coverage, which is often unavailable underground**
- C) Satellite connectivity automatically activates in tunnels
- D) Tunnel infrastructure includes embedded PC5 base stations

✓ **Correct Answer: B**

105. C-V2X latency 'under 10ms' — in which specific scenario would even 10ms be insufficient?

- A) Downloading HD maps
- B) Emergency braking at very high speed — at 200km/h, 10ms corresponds to 55cm of travel; future platooning with gaps under 5m may require sub-5ms latency**
- C) Traffic signal optimization
- D) Cloud-based vehicle diagnostics

✓ **Correct Answer: B**

106. 'Intelligent resource scheduling and sensing' improves spectrum efficiency in C-V2X. What would happen without this mechanism in a dense deployment?

- A) Vehicles would switch to DSRC
- B) Without resource coordination, all vehicles would randomly contend for the same resources simultaneously, causing destructive interference similar to DSRC congestion but without CSMA/CA's backoff mitigation**
- C) RSUs would centrally coordinate all transmissions
- D) Bandwidth would automatically increase to accommodate more users

✓ **Correct Answer: B**

107. A V2P (Vehicle-to-Pedestrian) application sends a collision alert to a pedestrian's phone. What specific technical capability must the pedestrian device have?

- A) A full OBU with GNSS
- B) A C-V2X-capable module (PC5 interface) or a V2N connection via Uu to receive the proximity alert — a standard smartphone without V2X capability cannot directly receive PC5 messages**
- C) The pedestrian's phone receives the alert via Bluetooth from nearby vehicles
- D) V2P alerts are only displayed on vehicle dashboards, not pedestrian devices

✓ **Correct Answer: B**

108. The 5G core manages 'mobility and session control' in V2X. What happens to a vehicle's V2X sessions during high-speed handover between cells?

- A) All V2X communication pauses during handover
- B) The 5G core (AMF/SMF) manages seamless handover — session continuity is maintained by migrating the UE context to the new cell with minimal interruption to ongoing V2X services**
- C) Vehicles must reconnect to V2X servers after each handover
- D) PC5 communication stops during Uu handover

✓ **Correct Answer: B**

109. C-V2X 'improves spectrum efficiency through intelligent resource scheduling.' How does this specifically differ from DSRC's approach?

- A) C-V2X uses more bandwidth
- B) C-V2X's sensing-based and scheduled resource allocation actively avoids resource conflicts, while DSRC's CSMA/CA is reactive — it mitigates but does not prevent collisions**
- C) C-V2X uses more transmit power to overcome interference
- D) DSRC uses intelligent scheduling while C-V2X uses random access

✓ **Correct Answer: B**

110. 'Cloud-assisted vehicular decision making' via the 5G core Uu interface refers to what specific capability?

- A) The cloud replaces the OBU decision-making processor
- B) Cloud services can aggregate data from many vehicles to provide context-aware services (HD maps, collective perception, predictive routing) that individual vehicles cannot compute alone**
- C) The cloud controls vehicle steering and braking remotely
- D) Decision making always occurs at the cloud, not in the vehicle

✓ **Correct Answer: B**

111. Why is 'coordinated driving and platooning' described as a key C-V2X urban mobility advantage rather than a DSRC advantage?

- A) DSRC does not support platooning
 - B) Platooning at close gaps (<10m) requires very reliable, low-latency, high-density-capable communication — C-V2X's superior congestion handling and reliability make tight platooning more feasible than DSRC**
 - C) DSRC platooning requires RSU approval
 - D) Platooning only works on highways, not in urban areas
- ✓ Correct Answer: B

112. In C-V2X, 'sidelink reliability' is described as critical for safety alerts. What would 'unreliable sidelink' mean in a real collision scenario?

- A) The infotainment system would restart
 - B) A vehicle's emergency brake alert transmitted via unreliable PC5 might not be received by following vehicles, eliminating any warning time and preventing automated safety responses**
 - C) The Uu interface automatically takes over
 - D) Unreliable sidelink would increase latency to 100ms
- ✓ Correct Answer: B

113. V2I use-case: 'Infrastructure sharing vehicle speed.' What specific safety benefit does an infrastructure knowing vehicle speeds provide?

- A) The infrastructure can display speed on road signs
 - B) Infrastructure aggregating multiple vehicles' speeds can detect dangerous speed differentials (e.g., sudden slowdowns), issue targeted warnings, and adapt signal timing in real-time**
 - C) RSUs use vehicle speed data for toll calculations
 - D) Vehicle speed data is stored for law enforcement
- ✓ Correct Answer: B

114. In C-V2X architecture, what does the 'GNSS receiver' specifically contribute that no other component provides?

- A) GNSS provides network connectivity
 - B) GNSS provides absolute vehicle position (lat/lon/elevation) with sub-meter accuracy — this position data is essential for BSMs and location-based V2X services and cannot be replaced by cellular or V2X signals alone**
 - C) GNSS manages certificate retrieval from SCMS
 - D) GNSS provides timing synchronization for MAC-layer scheduling
- ✓ Correct Answer: B

115. 'Real-time hazard map updates' benefit from V2N latency improvements. Why is V2N (not V2V) the correct path for map updates?

- A) V2V has insufficient range for map distribution
 - B) Hazard maps are generated by aggregating data from many vehicles at a cloud server, then distributed wide-area — this requires the V2N cloud connectivity of Uu, not direct PC5 links**
 - C) PC5 supports higher data rates than Uu for map transfers
 - D) Map updates are sent directly between vehicles via PC5 broadcast
- ✓ Correct Answer: B

116. The PC5 interface sends 'safety-critical broadcast messages.' What makes broadcast (rather than unicast) the correct choice for safety messages?

- A) Broadcast is simpler to implement

B) Safety events (collisions, hazards) are relevant to ALL nearby vehicles simultaneously — broadcast reaches all nodes with one transmission, whereas unicast would require N individual transmissions to N neighbors

C) Unicast requires certificates that are unavailable in DSRC

D) Broadcast is mandated by regulatory bodies for all V2X

✓ **Correct Answer: B**

117. Why does 'cloud-assisted traffic optimization' specifically require the Uu interface rather than PC5?

A) PC5 cannot transmit traffic data

B) Cloud optimization requires vehicles to send data to and receive from a remote cloud server — PC5 is limited to direct local device-to-device communication and cannot reach internet-based cloud services

C) The cloud only accepts Uu-formatted packets

D) Uu has higher data rates needed for optimization algorithms

✓ **Correct Answer: B**

118. 'C-V2X supports latency under 10ms' — this target applies to which specific path?

A) Uu interface through the cellular network

B) PC5 direct sidelink — the Uu path through the network typically has 30-100ms latency; only the direct PC5 link can achieve sub-10ms for V2V safety applications

C) The 10ms target applies equally to both PC5 and Uu

D) The 10ms target is for cloud processing time, not transmission

✓ **Correct Answer: B**

119. Non-safety application: 'remote vehicle diagnostics.' Why is this NOT suitable for PC5 direct communication?

A) PC5 cannot transmit diagnostic data

B) Remote diagnostics require connection to a distant dealer or OEM server — this requires internet connectivity via Uu, not direct local PC5 communication to nearby vehicles

C) Diagnostics require higher bandwidth than PC5 supports

D) Remote diagnostics are prohibited on PC5 by 3GPP standards

✓ **Correct Answer: B**

120. In C-V2X, what is the functional role of the RSU compared to its role in DSRC-only systems?

A) RSUs become unnecessary in C-V2X systems

B) In C-V2X, RSUs serve as V2I communication bridges supporting both PC5 direct communication and potentially Uu backhaul, while also running edge applications — broader than DSRC RSUs which primarily broadcast V2I safety messages

C) C-V2X RSUs only support V2N communication

D) RSU function is identical in both DSRC and C-V2X

✓ **Correct Answer: B**

Lecture 7 · Evolution to 5G-V2X and NR-V2X

121. 5G-V2X targets <1ms latency. What specific 5G NR PHY mechanism achieves sub-millisecond transmission compared to LTE's ~1ms subframe?

- A) Higher transmit power in 5G NR
- B) Mini-slot based transmission — 5G NR supports transmission durations as short as 2 OFDM symbols (vs. 14 in LTE), enabling sub-millisecond scheduling granularity**
- C) 5G NR uses a different frequency band with lower propagation delay
- D) 5G NR eliminates the guard periods between transmissions

✓ **Correct Answer: B**

122. NR-V2X Mode 2 uses 'distributed resource allocation.' What makes this superior to LTE-V2X Mode 4 sensing in specific scenarios?

- A) Mode 2 requires less power than Mode 4
- B) NR Mode 2 incorporates UE cooperative scheduling — vehicles can share resource selection information with neighbors, reducing collision probability compared to purely autonomous sensing**
- C) Mode 2 operates only via the Uu interface
- D) Mode 2 eliminates the need for GNSS synchronization

✓ **Correct Answer: B**

123. 'Cooperative perception' in 5G-V2X means vehicles share sensor data. What technical challenge does this create that basic BSM exchange does not?

- A) Security certificates are insufficient for sensor data
- B) Sensor data (camera frames, LIDAR point clouds) is orders of magnitude larger than BSMs, requiring high-throughput reliable transmission that exceeds LTE-V2X capabilities but is addressed by 5G NR's higher data rates**
- C) Cooperative perception requires cloud processing only
- D) GNSS accuracy is insufficient for sensor data sharing

✓ **Correct Answer: B**

124. URLLC requires 99.999% reliability. What specific 5G NR mechanism enhances reliability compared to LTE?

- A) Higher antenna count in 5G NR
- B) HARQ with very low retransmission rounds, redundant resource transmission (duplication), and link adaptation — collectively ensuring near-guaranteed delivery within tight latency bounds**
- C) 5G NR uses error-free quantum encryption
- D) 99.999% reliability is achieved by increasing transmit power

✓ **Correct Answer: B**

125. 'Grant-free access' in NR-V2X eliminates scheduling requests. What latency benefit does this provide?

- A) Grant-free access increases data rate
- B) Traditional LTE requires a vehicle to request resources and wait for a grant before transmitting — this request/grant round trip adds 5-20ms. Grant-free eliminates this, enabling immediate transmission**
- C) Grant-free access is only available via Uu interface
- D) Grant-free access replaces security certificates

✓ **Correct Answer: B**

126. 5G-V2X introduces 'dedicated sidelink control channels.' What specific improvement does this provide over LTE-V2X's sidelink control?

- A) Dedicated channels use higher frequency
- B) Dedicated sidelink control channels provide more reliable and frequent resource coordination feedback between neighboring vehicles, improving Mode 2 distributed scheduling effectiveness**
- C) Control channels enable BSMs to bypass security checks
- D) Dedicated control channels eliminate the need for GNSS synchronization

✓ **Correct Answer: B**

127. Why does 5G-V2X support 'unicast' over sidelink while LTE-V2X primarily supports broadcast only?

- A) Unicast is faster than broadcast in all scenarios
- B) Advanced use cases like cooperative driving and sensor sharing require targeted, acknowledged, point-to-point links between specific vehicles — broadcast is insufficient for these applications requiring reliability guarantees**
- C) Unicast reduces interference between vehicles
- D) Regulatory requirements mandated unicast for 5G sidelink

✓ **Correct Answer: B**

128. 'Teleoperated driving enhancements' are focused in Release 16. Why does teleoperation specifically require 5G-V2X rather than LTE-V2X?

- A) Teleoperation only works on 5G hardware
- B) A remote human operator controlling a vehicle requires sub-10ms round-trip latency (to feel responsive) and ultra-high reliability — LTE-V2X's 20ms latency makes teleoperation feel sluggish and unsafe**
- C) LTE-V2X does not support video transmission
- D) Teleoperation requires millimeter-wave frequencies only

✓ **Correct Answer: B**

129. 'Groupcast' addressing in NR-V2X serves a different function than both unicast and broadcast. What is its specific technical role?

- A) Groupcast is identical to multicast in IP networks
- B) Groupcast allows a message to target a specific defined group (e.g., a platoon) — more efficient than broadcast (sends to ALL) and more scalable than N unicasts, with optional HARQ feedback from group members**
- C) Groupcast is only used for entertainment content in vehicles
- D) Groupcast in NR-V2X requires RSU coordination for group formation

✓ **Correct Answer: B**

130. The V2X evolution path: DSRC → LTE-V2X → NR-V2X. At each step, which dimension improves MOST significantly?

- A) Coverage range increases most at each step
- B) Latency decreases and reliability increases at each generation — DSRC:~50ms/>90%, LTE-V2X:20ms/>90%, NR-V2X:<1ms/99.999% — enabling progressively more demanding autonomous driving applications**
- C) Data rate is the primary improvement at each step
- D) Security strength is the primary improvement at each step

✓ **Correct Answer: B**

131. Edge computing in 5G-V2X reduces latency specifically for which type of application?

A) Applications that store vehicle history logs

B) Applications requiring real-time analytics (cooperative perception, collision prediction) — by processing at an edge server near the vehicle rather than a distant cloud, round-trip time drops from 100-200ms to 1-5ms

C) Applications that download HD maps periodically

D) Applications that manage security certificate issuance

✓ Correct Answer: B

132. OFDMA (vs LTE-V2X's OFDM) in NR-V2X provides which specific multiple access advantage?

A) OFDMA uses different frequencies than OFDM

B) OFDMA allows different vehicles to transmit simultaneously on orthogonal subcarrier groups — improving spectrum efficiency and reducing inter-vehicle interference compared to OFDM where only one node transmits at a time in a slot

C) OFDMA eliminates the need for resource allocation

D) OFDMA is mandatory for safety messaging in 3GPP Release 16

✓ Correct Answer: B

133. NR-V2X introduces 'multicast over sidelink.' For vehicle platooning specifically, why is multicast superior to repeated unicast?

A) Multicast requires less security overhead

B) A platoon leader transmitting one multicast frame reaches all following vehicles simultaneously — N unicast transmissions would consume N× the spectrum resources and have N× higher collision probability

C) Unicast cannot be used in platooning scenarios

D) Multicast has lower latency than unicast per message

✓ Correct Answer: B

134. Why is 5G NR-V2X described as enabling 'cooperative driving' while LTE-V2X primarily enabled 'assisted driving'?

A) NR-V2X has more RSUs

B) Cooperative driving requires vehicles to share high-frequency sensor data and coordinate maneuvers in real-time (<1ms) — LTE-V2X's 20ms latency and lower reliability make true real-time maneuver coordination unsafe at high speeds

C) LTE-V2X does not support V2V communication

D) Cooperative driving requires satellite connectivity

✓ Correct Answer: B

135. 'Advanced sidelink features' in Release 16 specifically include what capability that Release 14 lacked?

A) PC5 interface support

B) Sidelink HARQ feedback in unicast mode — LTE-V2X sidelink was broadcast-only with no acknowledgment; NR sidelink supports unicast with HARQ, dramatically improving link reliability for targeted communications

C) Mode 4 autonomous resource selection

D) GNSS-based synchronization for sidelink

✓ Correct Answer: B

136. Why does NR-V2X specifically need 'dedicated sidelink synchronization signals' in addition to network-provided synchronization?

- A) To reduce OBU cost
- B) Out-of-coverage vehicles cannot receive cellular synchronization — dedicated sidelink synchronization allows vehicles to maintain timing alignment with each other using GNSS-derived or peer-to-peer synchronization**
- C) Network synchronization is insufficient for Uu interface timing
- D) Synchronization signals are used only for security purposes

✓ **Correct Answer: B**

137. 5G-V2X supports 'sensor information sharing' as one of four autonomous driving scenario groups. What makes raw sensor sharing qualitatively different from BSM sharing?

- A) Sensor data uses different encryption
- B) BSMs contain derived data (position, speed) — sensor sharing transmits raw perception data (camera frames, LIDAR clouds) enabling other vehicles to perceive the sender's environment, extending their effective sensing range beyond their own sensors**
- C) Sensor sharing is done via RSU relay only
- D) Sensor sharing replaces GPS with collaborative positioning

✓ **Correct Answer: B**

138. Release 16 focuses on 'teleoperated driving.' What does this use case require from the sidelink that no previous V2X generation could provide?

- A) Higher transmit power
- B) Ultra-reliable bidirectional real-time link: the remote operator needs <10ms round-trip control commands (downlink) AND real-time high-quality video from the vehicle (uplink) — NR-V2X URLLC provides this while LTE-V2X cannot**
- C) Satellite communication support
- D) No special sidelink requirement — teleoperation uses Uu only

✓ **Correct Answer: B**

139. In NR-V2X, 'resource allocation' involves both Mode 1 (network scheduled) and Mode 2 (autonomous). When would a fleet operator choose Mode 1 over Mode 2?

- A) Mode 1 is always preferred in dense environments
- B) Mode 1 is optimal when vehicles are in cellular coverage and coordinated resource management by the gNB can reduce inter-vehicle interference — especially for vehicles in a managed fleet with predictable communication patterns**
- C) Mode 1 is the only option for emergency vehicles
- D) Mode 1 is chosen when vehicles move faster than 100 km/h

✓ **Correct Answer: B**

140. 'Cooperative maneuvering messages' in NR-V2X differ from BSMs. What do they contain that BSMs do not?

- A) Vehicle identification numbers
- B) Intended future trajectory, planned maneuver (e.g., lane change in 2 seconds), and coordination requests — enabling neighboring vehicles to proactively adjust their behavior to accommodate the intended action**
- C) Engine status and fuel level
- D) Insurance and registration information

✓ **Correct Answer: B**

Lecture 8 · 5G-V2X Enhancement Technologies

141. UCNC (User-Centric No Cell Edge) solves what specific problem that traditional cellular handover cannot?

- A) UCNC eliminates the need for base stations
- B) Traditional handover creates coverage gaps and latency spikes at cell edges — UCNC creates virtual/hyper cells where vehicles are simultaneously served by multiple gNBs, eliminating cell-edge boundaries**
- C) UCNC reduces the number of cells needed
- D) UCNC is only relevant for pedestrians, not vehicles

✓ **Correct Answer: B**

142. Massive MIMO uses many antennas simultaneously. What specific 5G-V2X challenge does this address?

- A) Massive MIMO increases the number of V2X channels
- B) Massive MIMO enables spatial multiplexing — different vehicles in the same frequency-time resource can be separated using beam steering, dramatically increasing system capacity without additional spectrum**
- C) Massive MIMO reduces OBU manufacturing cost
- D) Massive MIMO increases the range of PC5 direct links

✓ **Correct Answer: B**

143. mmWave frequencies (>24 GHz) provide high bandwidth but face specific challenges for vehicular use. Which challenge is most fundamental?

- A) mmWave cannot support OFDM modulation
- B) mmWave has very short propagation range and is highly susceptible to blockage by buildings, trucks, and even rain — requiring precise beam alignment that is extremely challenging at vehicular speeds**
- C) mmWave is too expensive for vehicle deployment
- D) mmWave frequencies are reserved for military use only

✓ **Correct Answer: B**

144. Network slicing creates 'dedicated QoS behavior' for safety traffic. Without slicing, what specific failure mode can occur?

- A) Safety messages are automatically prioritized regardless
- B) A burst of infotainment traffic could temporarily consume network resources intended for safety messages, causing safety message delay — slicing guarantees isolated resource allocation**
- C) Network capacity is always sufficient without slicing
- D) Safety message traffic is always smaller than infotainment traffic

✓ **Correct Answer: B**

145. Dual connectivity allows simultaneous LTE and 5G connections. For V2X specifically, what does this enable?

- A) Vehicles can use two frequencies simultaneously for higher speed
- B) During 5G coverage gaps, LTE provides seamless fallback — vehicles maintain V2N connectivity and Uu-based services without interruption, ensuring service continuity during the transition period**
- C) Dual connectivity halves the latency of each interface
- D) Dual connectivity is only used for infotainment streaming

✓ Correct Answer: B

146. MEC processes data 'close to the vehicle.' What specifically is meant by 'close to' and why does distance matter?

- A) MEC is installed inside vehicles
- B) MEC servers are co-located with or near base stations — the key is reducing the geographic distance data must travel, since speed-of-light propagation over long distances contributes significantly to round-trip latency**
- C) MEC is located in the RSU hardware
- D) Distance matters only for security, not latency

✓ Correct Answer: B

147. The SMF in the 5G core 'supports multicast and QoS control.' In the context of V2X multicast delivery, what role does SMF play?

- A) SMF transmits the multicast packets to vehicles
- B) SMF establishes and manages the multicast session — assigning appropriate QoS parameters, selecting transmission paths, and coordinating between the gNB and UPF for efficient one-to-many delivery**
- C) SMF handles security certificate validation for multicast
- D) SMF is a radio frequency management function

✓ Correct Answer: B

148. Beamforming 'directs energy toward the intended receiver.' For fast-moving vehicles, what additional capability must the beamforming system have?

- A) Higher transmit power for moving targets
- B) Beam tracking — the beam must continuously follow the moving vehicle, requiring real-time angle estimation and beam steering updates faster than the vehicle moves out of the beam footprint**
- C) Beamforming is static once configured for a route
- D) Moving vehicles automatically return to broadcast mode

✓ Correct Answer: B

149. 'Unified QoS across NR-Uu and NR-PC5' solves what specific interoperability problem?

- A) It reduces the number of certificates needed
- B) A vehicle switching between PC5 direct communication and Uu cellular communication needs consistent QoS treatment — without unified QoS, the switch could cause service degradation or priority loss for safety messages**
- C) Unified QoS eliminates the need for network slicing
- D) It allows PC5 and Uu to share the same physical frequency

✓ Correct Answer: B

150. MEC is described as critical for 'cooperative perception.' What would happen without MEC at intersections with 100+ vehicles sharing sensor data?

- A) Sensor data would be sent to the cloud without issue
- B) Without edge processing, all sensor data from 100 vehicles would travel to the distant cloud for aggregation — the round-trip latency (100-300ms) would make the processed collective perception outdated by the time it returns**
- C) Vehicles would stop sharing sensor data automatically
- D) RSUs would buffer all sensor data locally without processing

✓ Correct Answer: B

151. URLLC's 99.999% reliability target corresponds to what maximum failure rate?

- A) 1 failure per 1,000 transmissions
- B) 1 failure per 100,000 transmissions — in a V2X system with 10 safety messages per second, this means approximately 1 message failure per ~2.8 hours per vehicle**
- C) 1 failure per 1 million transmissions
- D) 0 failures — URLLC means perfect reliability

✓ Correct Answer: B

152. 'Virtual cell' architecture creates a 'seamless wide-area service region.' How does this specifically differ from UCNC's 'hyper cell'?

- A) Virtual cells are smaller than hyper cells
- B) A virtual cell serves an individual vehicle from the most suitable gNB(s) — it's vehicle-centric. A hyper cell is the aggregated coverage region formed by multiple cells — it's infrastructure-centric. Together they eliminate edge effects**
- C) Virtual cells are only used indoors; hyper cells outdoors
- D) Virtual and hyper cells are identical — just different names

✓ Correct Answer: B

153. In NR-PC5, 'multicast and unicast support' was a major advancement over LTE-PC5. What technical requirement of the link does unicast with HARQ feedback introduce?

- A) Higher transmit power
- B) A feedback channel from the receiver to confirm (ACK) or request retransmission (NACK) — this requires knowing the target receiver's identity and establishing a unicast link, fundamentally changing PC5 from anonymous broadcast to addressed communication**
- C) A dedicated SCMS certificate for unicast links
- D) Unicast over PC5 requires RSU relay

✓ Correct Answer: B

154. 'Critical safety messages receive highest QoS priority' in the NR-Uu model. What happens at the gNB when both infotainment and safety messages queue simultaneously?

- A) Messages are served in arrival order (FIFO)
- B) The gNB's scheduler prioritizes the safety message for immediate transmission in the next available scheduling opportunity, potentially preempting or delaying the infotainment message — this is the function of QoS priority classes**
- C) Safety and infotainment messages are transmitted simultaneously on different channels
- D) The OBU decides which message to send first, not the gNB

✓ Correct Answer: B

155. Beamforming and Massive MIMO are described as 'improving wireless throughput and reliability' together. Why are they more powerful combined than individually?

- A) They operate at different frequencies, eliminating interference
- B) Massive MIMO provides spatial multiplexing (multiple simultaneous links) while beamforming focuses energy directionally — combined, they enable multiple high-SNR directional links simultaneously, multiplying both throughput and reliability**
- C) They use different security certificates, enabling parallel transmissions
- D) Combined, they reduce transmission power requirements

✓ Correct Answer: B

156. 5G enhancement technology 'flexible scheduling' is specifically valuable for V2X because vehicular traffic is:

- A) Always periodic and predictable
- B) Highly variable — dense intersections alternate with sparse highways; flexible dynamic scheduling adapts resource allocation to instantaneous demand rather than worst-case static allocation**
- C) Exclusively non-safety infotainment
- D) Bounded by DSRC channel allocation

✓ Correct Answer: B

157. Why is mmWave 'challenging for vehicular communication' specifically due to 'beam alignment'?

- A) mmWave beams are too wide to focus
- B) At 200km/h, a vehicle's angular position relative to a base station changes rapidly — maintaining a narrow mmWave beam precisely aligned with a fast-moving vehicle requires continuous real-time beam tracking with sub-millisecond update rates**
- C) Beam alignment requires satellite GPS not available in cities
- D) mmWave beam alignment is only needed for fixed RSUs

✓ Correct Answer: B

158. Network slicing provides 'separate QoS behavior for safety and infotainment traffic.' What would happen in an unsliced network during a traffic accident?

- A) Safety messages would automatically preempt infotainment traffic
- B) A surge of accident-related V2X safety messages could compete with ongoing infotainment traffic in the same unsliced network pool — without isolation guarantees, safety message delivery would depend on whatever resources remain after general traffic**
- C) Unsliced networks always give safety messages priority
- D) Infotainment traffic automatically pauses during accidents

✓ Correct Answer: B

159. MEC 'reduces reliance on central cloud servers.' For V2X cooperative perception specifically, why is central cloud reliance problematic even with high bandwidth?

- A) Cloud servers cannot process V2X data
- B) Round-trip propagation to a distant cloud server introduces 50-300ms latency — by the time processed collective perception returns to vehicles, moving objects may have changed position by several meters, making the data too stale for safety decisions**
- C) Central cloud servers are too expensive for V2X use
- D) Cloud servers cannot handle the encryption of V2X sensor data

✓ Correct Answer: B

160. UCNC's 'dual connectivity' means a vehicle connects to multiple gNBs simultaneously. What specific V2X benefit does this provide during handover?

- A) Dual connectivity doubles the data rate
- B) The vehicle maintains a connection to both the old and new gNB during handover — safety-critical V2X sessions are maintained without interruption, eliminating the brief disconnection that single-connection handover creates**
- C) Dual connectivity is required for URLLC by 3GPP standard
- D) Dual connectivity is only used for infotainment during handover

✓ **Correct Answer: B**

Lecture 9 · PC5/Uu Integration, Deployment, MEC, and RSUs

161. 'Sidelink-based positioning' is useful when GNSS is degraded. What specific technical mechanism does PC5 sidelink use for positioning?

- A) PC5 sidelink accesses GNSS satellites directly
- B) Vehicles measure signal time-of-arrival, round-trip time, or angle-of-arrival from neighboring UEs' sidelink signals to estimate relative distances and positions — creating a local positioning network independent of satellites**
- C) RSUs provide positioning signals when GNSS fails
- D) Sidelink positioning uses cellular tower triangulation

✓ Correct Answer: B

162. Why must new vehicles support BOTH LTE-V2X and 5G-V2X (dual mode)?

- A) Regulatory bodies mandate both modes
- B) During the transition period, the vehicle fleet will contain both legacy LTE-V2X vehicles and newer 5G-V2X vehicles — dual-mode vehicles can communicate with both, maintaining safety in mixed fleets**
- C) Dual mode is required only for platooning applications
- D) 5G-V2X alone is sufficient once deployed in an area

✓ Correct Answer: B

163. MEC at intersections specifically enables 'real-time signal coordination and collision avoidance.' What data does the intersection MEC process that individual vehicles cannot?

- A) Satellite GPS signals for all vehicles
- B) Data from ALL vehicles approaching the intersection simultaneously — individual vehicles only see their own trajectory, while MEC aggregates all trajectories to detect conflict points no individual vehicle can predict**
- C) Traffic light timing from the TMC
- D) Historical accident data for the intersection

✓ Correct Answer: B

164. 'Relay-capable UE operation' extends coverage for out-of-coverage vehicles. What specific technical challenge does the relay UE face?

- A) The relay UE must switch to satellite communication
- B) The relay UE must simultaneously maintain a direct PC5 link to the out-of-coverage vehicle AND a Uu link to the cellular network — requiring radio resource management for two simultaneous half-duplex links on potentially the same or different channels**
- C) Relay UEs automatically increase transmit power
- D) Relay UEs require a special license from the operator

✓ Correct Answer: B

165. Tunnel scenarios: 'PC5 direct links' are preferred. What happens to Uu-based V2N services in a long tunnel?

- A) Uu performance improves inside tunnels due to waveguide effect
- B) Cellular signal typically degrades or disappears in long tunnels without embedded antennas — V2N services become unavailable, but PC5 V2V communication continues between vehicles in the tunnel without infrastructure**
- C) Tunnels always have RSUs that maintain Uu connectivity
- D) Vehicles automatically switch to Wi-Fi inside tunnels

✓ **Correct Answer: B**

166. 'Full IoV ecosystem deployment' is the final deployment roadmap phase. What distinguishes this from initial C-V2X deployment?

- A) IoV uses different hardware than C-V2X
- B) Full IoV ecosystem integrates V2X with cloud services, AI analytics, smart city infrastructure, and diverse mobility platforms — not just vehicle-to-vehicle safety messaging but a comprehensive intelligent transportation system**
- C) IoV deployment eliminates RSUs entirely
- D) Full IoV is defined as 100% vehicle penetration rate

✓ **Correct Answer: B**

167. 'V2X T-Box acts as a central communication gateway.' What does this mean architecturally?

- A) T-Box replaces OBU in modern vehicles
- B) The T-Box aggregates in-vehicle network data (CAN bus, sensors) and manages external V2X communication interfaces (PC5, Uu, Wi-Fi), acting as the single integration point between internal vehicle systems and external V2X networks**
- C) T-Box is a cloud server not installed in vehicles
- D) T-Box only handles security certificate management

✓ **Correct Answer: B**

168. Combining NR-Uu and NR-PC5 provides 'complementary reliability and latency improvements.' What is the specific failure mode that each interface protects against?

- A) Both interfaces protect against the same failure modes
- B) PC5 protects against cellular outages (direct local communication works without infrastructure); Uu protects against PC5 range limitations and provides wide-area services — failures in one interface are compensated by the other**
- C) Uu protects against packet loss; PC5 protects against latency
- D) PC5 protects against security attacks; Uu protects against congestion

✓ **Correct Answer: B**

169. 'RSU/MEC-assisted V2X deployment' is described as best for intersection coordination. What specific technical advantage does RSU-mounted MEC have over cloud-based processing?

- A) RSU MEC has more processing power than cloud servers
- B) RSU-mounted MEC processes vehicle data locally at the intersection — eliminating the round-trip to a remote cloud server and providing sub-10ms response time for safety coordination impossible with cloud latency**
- C) RSU MEC eliminates the need for V2X certificates
- D) RSU MEC enables encryption not available in cloud

✓ **Correct Answer: B**

170. 'Uu-based deployment challenge: latency and broadcasting limitations.' What specific broadcasting limitation does Uu have compared to PC5?

- A) Uu cannot transmit more than 1 Mbps
- B) Uu requires messages to go through the cellular network infrastructure — true one-to-all local broadcast (as in PC5) is not native to Uu; achieving broadcast-like V2X safety messaging via Uu requires multicast groups or individual unicasts, adding complexity and latency**
- C) Uu broadcasting is limited to 100m range

D) Uu cannot carry BSM-formatted messages

✓ **Correct Answer: B**

171. The RSU dual interface 'Uu + PC5' allows what specific capability that single-interface RSUs cannot provide?

A) Higher transmit power in both directions

B) The RSU can receive direct PC5 V2I messages from nearby vehicles AND forward relevant data to the cellular network via Uu — bridging local V2X communication to wide-area infrastructure services like TMC coordination

C) Dual interface eliminates the need for security certificates

D) Single-interface RSUs can perform all functions using software upgrades

✓ **Correct Answer: B**

172. 'Lack of continuous coverage' is a 5G-Uu deployment limitation. For safety-critical V2X, what architectural solution addresses this gap?

A) Vehicles wait until coverage is restored

B) PC5 direct communication provides a coverage-independent safety fallback — when Uu is unavailable, V2V safety messages continue via PC5, ensuring safety communication persists in coverage gaps

C) Satellite backhaul fills all coverage gaps

D) Safety messages are cached and sent when coverage is restored

✓ **Correct Answer: B**

173. 'PC5-based deployment is ideal for forward collision warning.' Why specifically is PC5 better than Uu for this application?

A) PC5 has higher bandwidth for collision data

B) Forward collision warning requires sub-100ms end-to-end latency — Uu's network round-trip adds 30-100ms, while PC5's direct link achieves 1-5ms, providing 10-20x more reaction time in the critical pre-collision window

C) Uu does not support collision warning messages

D) PC5 encryption is stronger for safety-critical data

✓ **Correct Answer: B**

174. 'Remote driving and platooning benefit most from MEC-based deployment.' For remote driving specifically, what does MEC provide that cloud cannot?

A) MEC provides more storage for remote driving logs

B) Remote driving requires <10ms control loop latency — cloud round-trip is 50-300ms, making the vehicle unresponsive to operator control. MEC processes vehicle sensor data and control commands locally, achieving sub-10ms round trips

C) MEC provides higher resolution camera feeds

D) MEC is needed because remote driving requires IPv6

✓ **Correct Answer: B**

175. Sidelink positioning improves localization in GNSS-challenged environments. What GNSS limitation specifically motivates this?

A) GNSS cannot operate above 100 km/h

B) Urban canyon environments (tall buildings) and tunnels block satellite signals, causing GNSS position errors of 10-50m — sidelink ranging between nearby vehicles provides sub-meter relative positioning that complements degraded GNSS

C) GNSS has insufficient update rate for vehicular use

D) GNSS cannot function in the 5.9 GHz band

✓ **Correct Answer: B**

176. The final C-V2X deployment roadmap phase ('Full IoV') integrates V2X with IoT. What specific capability does IoT integration add that pure V2X cannot provide?

A) IoT provides higher bandwidth than V2X

B) IoT integration connects vehicles to non-vehicle infrastructure data — traffic sensors, environmental monitors, smart traffic lights, pedestrian detection systems — creating situational awareness from data sources beyond vehicles and RSUs alone

C) IoT replaces SCMS for certificate management

D) IoT provides higher security than V2X

✓ **Correct Answer: B**

177. 'MEC deployment primarily supports autonomous driving and smart mobility.' What specific real-time analytics function does MEC perform that enables autonomous driving support?

A) MEC stores maps for offline autonomous driving

B) MEC aggregates V2X data from multiple vehicles/sensors in real-time to generate a collective environmental model (birds-eye view, pedestrian tracking, intersection state), then distributes this enhanced perception to all participating vehicles

C) MEC controls vehicle actuators directly

D) MEC monitors vehicle software for security threats

✓ **Correct Answer: B**

178. In the V2X deployment context, what does 'backward compatibility' between LTE-V2X and NR-V2X specifically require?

A) NR-V2X vehicles must use LTE hardware

B) NR-V2X vehicles must be able to communicate with LTE-V2X vehicles using common message formats and compatible PC5 interface modes — ensuring safety-critical V2V communication works regardless of whether a neighboring vehicle supports LTE or NR sidelink

C) LTE-V2X must be disabled when NR-V2X is available

D) Backward compatibility only applies to Uu, not PC5

✓ **Correct Answer: B**

179. An RSU 'collects local road data and forwards it to nearby edge computing resources.' What is the technical difference between 'co-located' MEC (at RSU) vs 'nearby' MEC (at base station)?

A) Co-located MEC is more expensive

B) Co-located MEC at the RSU eliminates backhaul latency between RSU and MEC — data processing occurs at the point of collection. Nearby MEC at the base station adds backhaul delay (typically 1-5ms) but benefits from more powerful shared infrastructure

C) Co-located MEC cannot process video data

D) RSU-co-located MEC and base-station MEC have identical performance

✓ **Correct Answer: B**

180. 'No infrastructure dependency' is listed as a key benefit of sidelink-based positioning. In what emergency scenario is this benefit most critical?

A) During rush hour traffic congestion

B) After a natural disaster or infrastructure failure that destroys RSUs and GPS ground station infrastructure — vehicles can still determine relative positions to each other using PC5 sidelink ranging, enabling coordinated emergency response and safe maneuvering

C) When OBU battery is low

D) When cellular coverage is at maximum capacity

✓ **Correct Answer: B**

Lecture 10 · Security Issues and Challenges in V2X Communication

181. A Sybil attack in V2X creates multiple fake identities. Why is this attack specifically dangerous for safety applications?

- A) Sybil attacks drain vehicle batteries
- B) Fake vehicle identities can generate false sensor data suggesting phantom traffic congestion, accidents, or road blockages — causing real vehicles to make dangerous maneuvers or emergency stops based on non-existent hazards**
- C) Sybil attacks only affect infotainment systems
- D) Sybil attackers can read encrypted V2X messages

✓ **Correct Answer: B**

182. Pseudonym certificates change frequently. What specific cryptographic property prevents an adversary from linking old and new pseudonyms?

- A) Pseudonyms use the same private key but different public keys
- B) Each pseudonym uses a completely independent key pair (private/public) with no mathematical relationship — proving the same vehicle owns two pseudonyms requires breaking cryptographic assumptions, not just observing message patterns**
- C) SCMS keeps pseudonym changes secret
- D) Pseudonym changes are masked by encrypting the vehicle's real IP address

✓ **Correct Answer: B**

183. 'Forward secrecy' protects past communications if a key is compromised. In a shared group key scenario (e.g., platoon), why is forward secrecy especially critical?

- A) Forward secrecy is less important for groups than individuals
- B) In a group with a shared key, if the key is compromised, ALL past group communications are exposed simultaneously — without forward secrecy, one compromised vehicle retroactively reveals the entire platoon's communication history**
- C) Forward secrecy only protects against replay attacks
- D) Group keys are always short-lived, providing inherent forward secrecy

✓ **Correct Answer: B**

184. CRLs (Certificate Revocation Lists) are distributed to invalidate compromised certificates. What V2X-specific challenge makes CRL distribution especially difficult?

- A) Vehicles cannot store CRL data
- B) CRLs must reach millions of vehicles in real-time or near-real-time — a revoked certificate is useless if vehicles haven't received the update, but distributing large CRLs to all vehicles over V2X bandwidth is a significant scalability challenge**
- C) CRLs are only needed for RSUs, not vehicles
- D) Regulatory rules prevent CRL distribution in some countries

✓ **Correct Answer: B**

185. A wormhole attack makes far vehicles appear close. Why is this attack specifically dangerous for cooperative driving applications?

- A) Wormhole attacks consume vehicle battery
- B) Cooperative driving decisions (e.g., whether to yield to a vehicle) depend on believing neighbors are physically nearby — a wormhole makes a distant vehicle appear adjacent, causing vehicles to yield to or coordinate with vehicles they will never physically encounter**

- C) Wormhole attacks only affect GPS accuracy
- D) Wormhole attackers can modify message content

✓ **Correct Answer: B**

186. V2X uses public key cryptography for authentication. What specific challenge of PKI does the V2X environment create that traditional internet PKI does not face?

- A) Vehicles cannot store private keys
- B) V2X requires verifying hundreds of BSMs per second from hundreds of vehicles simultaneously — the computational overhead of full PKI verification at this rate strains OBU processors, requiring optimized batch verification or implicit certificates**
- C) PKI cannot work with DSRC
- D) Public keys cannot be transmitted in DSRC BSMs

✓ **Correct Answer: B**

187. SCMS distributes pseudonym certificates with limited lifetime. Why is 'limited lifetime' specifically important beyond just privacy?

- A) Limited lifetime reduces SCMS storage requirements
- B) Short-lived certificates limit the window of opportunity for a compromised certificate to be used — if a vehicle is compromised, its certificates expire quickly, reducing the damage an attacker can cause using stolen credentials**
- C) Certificate lifetime is determined by vehicle speed
- D) Limited lifetime certificates use less computational resources

✓ **Correct Answer: B**

188. 'Non-repudiation' via digital signatures means a sender cannot deny sending a message. Why must V2X non-repudiation be balanced against privacy?

- A) Non-repudiation requires exposing the vehicle's permanent identity, which conflicts directly with privacy requirements — pseudonym systems weaken non-repudiation for regular communication while SCMS preserves accountability for misbehavior investigations by authorities only**
- B) Non-repudiation makes V2X messages larger
- C) Non-repudiation requires cellular connectivity
- D) Non-repudiation only applies to RSU messages, not vehicle BSMs

✓ **Correct Answer: A**

189. A DoS attack floods the V2X network. How does this differ from a jamming attack in terms of attacker capability required?

- A) DoS requires less sophisticated equipment than jamming
- B) Jamming requires only a radio transmitter capable of flooding the 5.9 GHz band with noise — available cheaply. DoS via message flooding requires a valid V2X certificate (or many Sybil identities) to inject authenticated messages at high rate — higher cryptographic access is needed**
- C) Both attacks require identical attacker capabilities
- D) DoS attacks are only possible via the internet, not radio

✓ **Correct Answer: B**

190. 'Linkability of vehicle messages' is a privacy concern. What specific information in V2X messages enables linkability even without knowing the real identity?

- A) Vehicle speed changes over time

B) Consistent pseudonyms used across many messages allow observers to track vehicle movement patterns — even without knowing the real identity, location history and behavior patterns reveal private information about the driver

C) Encryption certificates link messages automatically

D) Linkability is only possible if the attacker knows the vehicle's MAC address

✓ **Correct Answer: B**

191. MITM attack in V2X compromises 'message authenticity and confidentiality.' In DSRC broadcast mode, which of these two properties is harder to violate?

A) Authenticity is harder to violate

B) Confidentiality — DSRC BSMs are not encrypted (confidentiality is not required for safety broadcasts), making eavesdropping trivial. Authenticity is protected by digital signatures, making modification detectable (though not impossible for resource-rich adversaries)

C) Both are equally easy to violate

D) Neither can be violated in DSRC due to physical layer security

✓ **Correct Answer: B**

192. The SCMS 'handles misbehaving nodes by revoking certificates.' What is the specific technical process after misbehavior is detected?

A) The SCMS disables the vehicle's OBU remotely

B) The SCMS adds the vehicle's pseudonym certificates to the CRL, which is distributed to all V2X participants — subsequent messages from revoked certificates are rejected by receiving vehicles after they update their CRL

C) The SCMS notifies law enforcement automatically

D) Certificate revocation stops the vehicle from driving

✓ **Correct Answer: B**

193. 'Dynamic key distribution and revocation in real time' is the main challenge in V2X key management. What specific property of V2X makes traditional PKI key management insufficient?

A) V2X uses more certificates than other systems

B) Traditional PKI manages certificates for relatively static users (websites, users) — V2X must manage millions of rapidly changing pseudonym certificates for fast-moving vehicles while distributing revocation information quickly enough to prevent compromised certificates from being used

C) V2X security requirements are less strict than internet security

D) Traditional PKI is too expensive for V2X deployment

✓ **Correct Answer: B**

194. Message authentication 'confirms sender identity' — but in V2X with pseudonyms, what identity is actually being confirmed?

A) The vehicle's license plate identity

B) The pseudonym identity — authentication confirms the message was created by the holder of the private key corresponding to the pseudonym certificate, NOT the real-world vehicle identity. Real identity is hidden and only revealable by SCMS for misbehavior investigations

C) The driver's identity via biometric certificate

D) The vehicle's insurance and registration identity

✓ **Correct Answer: B**

195. A V2X attacker 'modifies a legitimate message' — which security property is violated and why does the receiver detect this?

A) Confidentiality is violated, detected by decryption failure

B) Integrity is violated — the message's digital signature is computed over the original content using the sender's private key; any modification of message bits causes the signature verification (using the public key) to fail, indicating tampering

C) Availability is violated, detected by timeout

D) Authentication is violated, detected by certificate mismatch

✓ **Correct Answer: B**

196. 'Short-lived certificates enhance pseudonymity.' What specific tracking resistance does short lifetime provide that a long-lived pseudonym does not?

A) Short-lived certificates use shorter encryption keys

B) A long-lived pseudonym used repeatedly over weeks creates a trackable profile — observing the same pseudonym at location A and B allows trajectory reconstruction. Short-lived certificates changed frequently limit the temporal window a single pseudonym can be observed, fragmenting tracking attempts

C) Short-lived certificates cannot be added to CRLs

D) Long-lived pseudonyms are prohibited by 3GPP standards

✓ **Correct Answer: B**

197. 'Latency is a security concern in V2X' because delays undermine timely threat response. What specific scenario demonstrates this?

A) An OBU that processes certificates slowly

B) A revocation notice reaching a vehicle 500ms after a misbehaving neighbor transmits a hazard spoofing message — the victim vehicle has already acted on the false data before the revocation arrives, illustrating how latency in security infrastructure enables attacks

C) A cellular network with high ping time

D) An RSU that validates certificates slowly

✓ **Correct Answer: B**

198. Jamming attacks 'isolate vehicles from the V2X network.' Against what type of V2X safety application is jamming MOST dangerous?

A) Navigation and mapping applications

B) Event-driven safety alerts (emergency braking, collision warnings) — these are non-periodic and critical; a vehicle being jammed at the exact moment it needs to receive an emergency alert misses the only warning with potentially fatal consequences

C) Weather updates and traffic information

D) Over-the-air software update applications

✓ **Correct Answer: B**

199. Asymmetric cryptography (public/private key) is chosen over symmetric for V2X authentication. What specific V2X operational requirement makes symmetric key approaches unscalable?

A) Symmetric keys are easier to intercept over radio

B) Symmetric authentication requires the verifier to share a secret key with every possible sender — in V2X, a vehicle must authenticate BSMs from any of millions of unknown nearby vehicles, requiring millions of pre-shared keys. Asymmetric keys allow authentication from any vehicle whose certificate is signed by a trusted CA

- C) Symmetric encryption is not supported by DSRC hardware
- D) SCMS cannot distribute symmetric keys

✓ **Correct Answer: B**

200. 'Providing scalable certificate distribution and revocation' is SCMS's key motivation. What scale specifically must SCMS handle?

- A) Thousands of certificates per city
- B) Hundreds of millions of pseudonym certificates for tens of millions of vehicles, each using multiple short-lived pseudonyms simultaneously — plus real-time revocation distribution to all active vehicles globally. No traditional PKI was designed for this scale**
- C) Certificate distribution only for RSUs and TMCs
- D) SCMS only handles certificates for emergency vehicles

✓ **Correct Answer: B**